Neutrons in Space

NE 630 | Nuclear Reactor Theory

31 LESSON

Read: FNRP 6.1–6.2.

Students will be able to show that neutrons are conserved (or not!) in a spatial volume given the neutron flux and appropriate cross-section data.

These notes represent a condensed summary of D&H, Ch. 4 with connections to Ch. 6 of FNRP and the diffusion equation. Equations numbered like (4–8) are from D&H, but I've modified nomenclature slightly in some places to match what we've previously adopted.

1 Characterizing Neutron Fields

Let the **angular density** of neutrons be described by

$$n(\mathbf{r}, E, \hat{\Omega}, t) \quad [\mathrm{cm}^{-3}\,\mathrm{eV}^{-1}\,\mathrm{sr}^{-1}], \tag{4–8}$$

so that $n(\mathbf{r}, E, \hat{\Omega}, t)\ d^3\mathbf{r}\ dE\ d\hat{\Omega}$ is the expected number of neutrons in the differential *phase-space* volume centered at $(\mathbf{r}, E, \hat{\Omega})$ at time t. The direction unit vector is $\hat{\Omega} = \mathbf{v}/\|\mathbf{v}\|$ where $\mathbf{v}$ is the neutron velocity.

The **angular flux** and **angular current** are

$$\psi(\mathbf{r}, E, \hat{\Omega}, t) = v\, n(\mathbf{r}, E, \hat{\Omega}, t) \quad [\mathrm{cm}^{-2}\,\mathrm{eV}^{-1}\,\mathrm{sr}^{-1}\,\mathrm{s}^{-1}], \tag{4–9}$$

and

$$\mathbf{j}(\mathbf{r}, E, \hat{\Omega}, t) = \hat{\Omega}\,\psi(\mathbf{r}, E, \hat{\Omega}, t) \quad [\mathrm{cm}^{-2}\,\mathrm{eV}^{-1}\,\mathrm{sr}^{-1}\,\mathrm{s}^{-1}], \tag{4–10}$$

where $v = \|\mathbf{v}\|$ is the neutron speed.

2 Scalar Quantities

Integrating over all angles gives the **scalar density** and **scalar flux**

$$n(\mathbf{r}, E, t) = \int_{4\pi} n(\mathbf{r}, E, \hat{\Omega}, t)\ d\hat{\Omega}, \tag{4–14}$$

$$\phi(\mathbf{r}, E, t) = \int_{4\pi} \psi(\mathbf{r}, E, \hat{\Omega}, t)\ d\hat{\Omega}. \tag{4–15}$$

The **current density** (or just **current**) is

$$\mathbf{J}(\mathbf{r}, E, t) = \int_{4\pi} \mathbf{j}(\mathbf{r}, E, \hat{\Omega}, t)\ d\hat{\Omega}, \tag{4–19}$$

so that $\mathbf{J}\cdot d\mathbf{A}$ is the net rate at which neutrons pass through a surface element $d\mathbf{A}$ per unit energy at time t. Although $\mathbf{J}$ and ϕ have the same units, $\mathbf{J}$ is a vector and represents net flow.

3 Integration over 4π

Integrals of the form

$$\int_{4\pi} \ldots\, d\hat{\Omega}$$

are sometimes hard to imagine.

An explicit example, in *spherical coordinates*, might be this solid-angle integral:

$$\int_{4\pi} f(\hat{\Omega})\ d\hat{\Omega} = \int_0^{2\pi}\int_0^{\pi} f(\theta,\phi)\ \sin\theta\, d\theta\, d\phi.$$

4 The Balancing Act

Consider some volume V with boundary surface S and outward normal $d\mathbf{s}$. The total number of neutrons in V with energies in $(E, E{+}dE)$ and directions in $d\hat{\Omega}$ about $\hat{\Omega}$ is

$$N(t) = \int_V n(\mathbf{r}, E, \hat{\Omega}, t)\ d^3\mathbf{r}\ dE\ d\hat{\Omega}.$$

If V does not depend on time, then differentiating $N(t)$ with respect to time gives

$$\int_V \frac{\partial n}{\partial t}(\mathbf{r}, E, \hat{\Omega}, t)\ d^3\mathbf{r}\ dE\ d\hat{\Omega} = \textbf{gains} - \textbf{losses}. \tag{4–24/25}$$

(1) any sources of n's
(2) n's "stream" into V through S
(3) n's scatter from $(E', \hat{\Omega}')$ to $(E, \hat{\Omega})$
(4) n's "stream" out of V through S
(5) n's that experience *any* interaction.

5 Inside the Volume

Let $s(\mathbf{r}, E, \hat{\Omega}, t)$ be a source density, so that $s(\cdots)\ d^3\mathbf{r}\ dE\ d\hat{\Omega}$ is the number of neutrons added to the volume per second. Then

$$(1) = \left[\int_V s(\mathbf{r}, E, \hat{\Omega}, t)\ d^3\mathbf{r}\right] dE\ d\hat{\Omega}. \tag{4–26}$$

Given $\Sigma_t(\mathbf{r}, E)$, the total interaction rate in the volume is

$$(5) = \left[\int_V \Sigma_t(\mathbf{r}, E)\,\psi(\mathbf{r}, E, \hat{\Omega}, t)\ d^3\mathbf{r}\right] dE\ d\hat{\Omega}. \tag{4–29}$$

This assumes an *isotropic* medium in which Σ_t does *not* depend on $\hat{\Omega}$. For the remaining balance terms, divide by $dE\ d\hat{\Omega}$ and work per unit energy and solid angle. Finally, *in-scattering* from $(E', \hat{\Omega}')$ to $(E, \hat{\Omega})$ contributes

$$(3) = \int_V \left[\int_0^{\infty}\int_{4\pi} \Sigma_s(\mathbf{r}, E' \to E, \hat{\Omega}' \to \hat{\Omega})\,\psi(\mathbf{r}, E', \hat{\Omega}', t)\, d\hat{\Omega}'\, dE'\right] d^3\mathbf{r}. \tag{4–31}$$

6 Through the Surface

The rate at which neutrons are gained or lost through a surface element is

$$\mathbf{j}(\mathbf{r}, E, \hat{\Omega}, t)\cdot d\mathbf{S} = \hat{\Omega}\,\psi(\mathbf{r}, E, \hat{\Omega}, t)\cdot d\mathbf{S}. \tag{4–32}$$

Integrating over the entire surface, the *net* number *leaving* is

$$(4) - (2) = \int_S \hat{\Omega}\,\psi(\mathbf{r}, E, \hat{\Omega}, t)\cdot d\mathbf{S}. \tag{4–33}$$

Recall Gauss' (divergence) theorem, i.e.,

$$\int_S \mathbf{F}(\mathbf{r})\cdot d\mathbf{S} = \int_V \nabla\cdot\mathbf{F}(\mathbf{r})\ d^3\mathbf{r}, \tag{4–34}$$

which, because $\hat{\Omega}$ is independent of $\mathbf{r}$, lets us write

$$(4) - (2) = \int_S \hat{\Omega}\psi\cdot d\mathbf{S} = \int_V \nabla\cdot\left(\hat{\Omega}\,\psi\right) d^3\mathbf{r} = \int_V \hat{\Omega}\cdot\nabla\psi\ d^3\mathbf{r}.$$

7 Shrinking V for Differential Balance

If V is arbitrary, then we collect the terms, take $V \to 0$ to eliminate the integrals, and welcome our new friend, the **neutron transport equation**:

$$\frac{1}{v}\frac{\partial\psi}{\partial t} + \hat{\Omega}\cdot\nabla\psi + \Sigma_t(\mathbf{r}, E)\,\psi(\mathbf{r}, E, \hat{\Omega}, t) = \int_0^{\infty}\int_{4\pi} \Sigma_s(\mathbf{r}, E' \to E, \hat{\Omega}' \to \hat{\Omega})\,\psi(\mathbf{r}, E', \hat{\Omega}', t)\ d\hat{\Omega}'\ dE' + s(\mathbf{r}, E, \hat{\Omega}, t). \tag{4–40}$$

Food for thought: what term would we add to include multiplication (fission)?

8 Eliminating $\hat{\Omega}$

While scary, the various terms comprising the NTE are familiar enough, but the dependence on $\hat{\Omega}$ remains a challenge for another day. Integrating the NTE over 4π leads to

$$\frac{1}{v}\frac{\partial\phi}{\partial t} + \nabla\cdot\mathbf{J}(\mathbf{r},E,t) + \Sigma_t(\mathbf{r},E)\,\phi(\mathbf{r},E,t) = \int_0^\infty \Sigma_s(\mathbf{r},E'\to E)\,\phi(\mathbf{r},E',t)\;dE' + S(\mathbf{r},E,t)\,, \tag{4–79}$$

which is called the **neutron continuity equation** (or NCE).

Ponderable. Have we simplified our life? Unfortunately, we went from the NTE with one unknown (ψ) to the new NCE with two unknowns (ϕ and $\mathbf{J}$)!

9 Finding Closure

We can't really do much with one equation and the unknowns ϕ and $\mathbf{J}$. The fix is a simple one: assume that the flow points where the neutron density, or, rather, flux falls:

$$\mathbf{J} \propto -\nabla\phi\,.$$

Ponderable: How do we go from $\propto$ to $=$?

Let D be that constant of proportionality so that

$$\mathbf{J} = -D\nabla\phi\,, \tag{FNRP 6.11}$$

which is known as **Fick's law**. Substitution of Fick's law into the NCE yields

$$\frac{1}{v}\frac{\partial\phi}{\partial t} - \nabla\cdot D(\mathbf{r},E)\nabla\phi(\mathbf{r},E,t) + \Sigma_t(\mathbf{r},E)\,\phi(\mathbf{r},E,t) = \int_0^\infty \Sigma_s(\mathbf{r},E'\to E)\,\phi(\mathbf{r},E',t)\;dE' + S(\mathbf{r},E,t)\,,$$

which is, of course, a long lost friend!

10 Simplify, Simplify

Now that we have space, it's time to say goodbye to t and to replace E by g:

$$-\nabla\cdot D_g(\mathbf{r})\nabla\phi_g(\mathbf{r}) + \Sigma_{tg}(\mathbf{r})\,\phi_g(\mathbf{r}) = \sum_{g'=1}^{G}\Sigma_{sg\leftarrow g'}\phi_{g'}(\mathbf{r}) + S_g(\mathbf{r})\,.$$

Example: For $G = 2$ and no upscatter, show that

$$-\nabla\cdot D_1(\mathbf{r})\nabla\phi_1(\mathbf{r}) + \Sigma_{r,1}(\mathbf{r})\,\phi_1(\mathbf{r}) = S_1(\mathbf{r})$$
$$-\nabla\cdot D_2(\mathbf{r})\nabla\phi_2(\mathbf{r}) + \Sigma_{a,2}(\mathbf{r})\,\phi_2(\mathbf{r}) = \Sigma_{s,2\leftarrow 1}\phi_1(\mathbf{r}) + S_2(\mathbf{r})$$

What if $G = 1$ and we introduce multiplication?

$$\boxed{-\nabla\cdot D(\mathbf{r})\nabla\phi(\mathbf{r}) + \Sigma_a(\mathbf{r})\,\phi(\mathbf{r}) = S(\mathbf{r}) + \bar{\nu}\Sigma_f(\mathbf{r})\,\phi(\mathbf{r})} \tag{FNRP 6.12}$$

11 Onto Slab Geometry

Much like integration over 4π hides some detail, so, too, does $\nabla\cdot D(\mathbf{r})\nabla\phi(\mathbf{r})$. Recall that, for Cartesian (or xyz) coordinates, the gradient and divergence operators act on scalar and vector functions according to

$$\nabla\phi(\mathbf{r}) = \mathbf{i}\frac{\partial\phi}{\partial x} + \mathbf{j}\frac{\partial\phi}{\partial y} + \mathbf{k}\frac{\partial\phi}{\partial z} \quad\text{and}\quad \nabla\cdot\mathbf{J}(\mathbf{r}) = \frac{\partial J_x}{\partial x} + \frac{\partial J_y}{\partial y} + \frac{\partial J_z}{\partial z}\,,$$

so that

$$\nabla\cdot\big(D(\mathbf{r})\,\nabla\phi(\mathbf{r})\big) = \frac{\partial}{\partial x}\left(D\frac{\partial\phi}{\partial x}\right) + \frac{\partial}{\partial y}\left(D\frac{\partial\phi}{\partial y}\right) + \frac{\partial}{\partial z}\left(D\frac{\partial\phi}{\partial z}\right)$$
$$= D(\mathbf{r})\,\nabla^2\phi(\mathbf{r}) + \big(\nabla D(\mathbf{r})\big)\cdot\big(\nabla\phi(\mathbf{r})\big).$$

If we make the bold assumption that $\phi(\mathbf{r}) = \phi(x)$, we find

$$\boxed{-\frac{d}{dx}\left(D(x)\frac{d\phi}{dx}\right) + \Sigma_a(x)\,\phi(x) = \bar{\nu}\Sigma_f(x)\,\phi(x) + S(x)}\,,$$

the one-speed diffusion equation in slab geometry.